Light

GCSE Computer Science – Data Representation: Media (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

How many bits does standard ASCII use to represent each character?

- **A.** 4 bits
- **B.** 7 bits
- **C.** 8 bits
- **D.** 16 bits

Q2 (1 mark)

What does the term 'pixel' stand for?

- **A.** Picture element

- B. Picture cell
- C. Pixel element
- D. Point element

Q3 (1 mark)

Sound waves in the real world are which type of signal?

- A. Binary
- B. Digital
- C. Analogue
- D. Hexadecimal

Q4 (1 mark)

What is a character set?

- A. A collection of characters that a computer can recognise and use
- B. A font style used for display
- C. A type of keyboard layout
- D. A method of encrypting text files

Q5 (1 mark)

What is colour depth in a bitmap image?

- ○ **A.** The number of colours in the image
- ○ **B.** The size of the image file
- ○ **C.** The brightness of the image
- ○ **D.** The number of bits used to represent each pixel

Q6 (1 mark)

What is sampling rate in digital audio?

- ○ **A.** The number of samples taken per second
- ○ **B.** The volume of the sound
- ○ **C.** The length of the audio file
- ○ **D.** The quality of the speakers

Q7 (1 mark)

What is a key advantage of Unicode over ASCII?

- ○ **A.** Unicode files are smaller
- ○ **B.** Unicode uses fewer bits per character
- ○ **C.** Unicode is faster to process

○ **D.** Unicode can represent characters from many different languages and alphabets

Q8 (1 mark)

How is a bitmap image constructed?

○ **A.** Using mathematical equations

○ **B.** Using text characters

○ **C.** Using lines and shapes

○ **D.** Using a grid of pixels, each with a colour value

Q9 (1 mark)

What is sample resolution (bit depth) in digital audio?

○ **A.** The number of bits used to store each sample

○ **B.** The frequency of the sound

○ **C.** The total file size

○ **D.** The number of audio channels

Q10 (1 mark)

In ASCII, if the code for 'A' is 65, what is the code for 'C'?

○ **A.** 66

- B. 97
- C. 68
- D. 67

Q11 (1 mark)

How is the size of a bitmap image typically expressed?

- A. In width x height pixels
- B. In megabytes only
- C. In centimetres
- D. In dots per inch

Q12 (1 mark)

What do Unicode and ASCII have in common for character codes 0-127?

- A. They use different codes for the same characters
- B. Only ASCII uses codes in this range
- C. Neither system uses these codes
- D. Unicode uses the same codes as ASCII for these characters

Q13 (2 marks)

Calculate the file size in bytes for a bitmap image that is 100 x 100 pixels with a colour depth of 8 bits.

- ○ **A.** 800 bytes
- ○ **B.** 8,000 bytes
- ○ **C.** 10,000 bytes
- ○ **D.** 80,000 bytes

Q14 (2 marks)

Calculate the file size in bits for a 10-second audio clip sampled at 1,000 Hz with a sample resolution of 8 bits.

- ○ **A.** 8,000 bits
- ○ **B.** 10,000 bits
- ○ **C.** 80,000 bits
- ○ **D.** 800,000 bits

Q15 (2 marks)

In ASCII, the code for 'a' (lowercase) is 97. What is the ASCII code for 'f'?

- ○ **A.** 102
- ○ **B.** 101
- ○ **C.** 103

- ○ **D.** 70

Q16 (2 marks)

An image uses a colour depth of 4 bits. How many different colours can be displayed?

- ○ **A.** 4 colours
- ○ **B.** 8 colours
- ○ **C.** 16 colours
- ○ **D.** 256 colours

Q17 (2 marks)

What happens to the file size when the sampling rate of an audio recording is doubled?

- ○ **A.** File size stays the same
- ○ **B.** File size doubles
- ○ **C.** File size halves
- ○ **D.** File size quadruples

Q18 (2 marks)

A 1-bit black and white image has the following row of binary data: 10110010. How many pixels does this row contain?

- ○ **A.** 4 pixels

- ○ **B.** 6 pixels
- ○ **C.** 8 pixels
- ○ **D.** 16 pixels

Q19 (2 marks)

Why was Unicode developed when ASCII already existed?

- ○ **A.** To make files smaller
- ○ **B.** To replace numbers with letters
- ○ **C.** To speed up text processing
- ○ **D.** To represent characters from many different alphabets and writing systems

Q20 (2 marks)

What effect does increasing the colour depth have on a bitmap image?

- ○ **A.** The image has fewer pixels
- ○ **B.** The image can display more colours but the file size increases
- ○ **C.** The image dimensions become larger
- ○ **D.** The image can be edited more easily

Q21 (2 marks)

What is a sample in digital audio?

- ○ **A.** A complete song
- ○ **B.** The total length of the recording
- ○ **C.** A type of audio file format
- ○ **D.** A measurement of the sound wave amplitude at a specific point in time

Q22 (2 marks)

An image is 640 x 480 pixels. How many pixels does the image contain in total?

- ○ **A.** 1,120 pixels
- ○ **B.** 30,720 pixels
- ○ **C.** 307,200 pixels
- ○ **D.** 3,072,000 pixels

Q23 (2 marks)

In ASCII, 'A' = 65 and 'a' = 97. What is the difference between uppercase and lowercase letter codes?

- ○ **A.** 16
- ○ **B.** 26
- ○ **C.** 32

- **D.** 64

Q24 (3 marks)

Calculate the file size in kilobytes (kB) for a bitmap image that is 800 x 600 pixels with a colour depth of 24 bits.

- **A.** 1,440 kB
- **B.** 14,400 kB
- **C.** 11,520 kB
- **D.** 1,152 kB

Q25 (3 marks)

Calculate the file size in bytes for a 30-second stereo audio clip sampled at 44,100 Hz with 16 bits per sample. Stereo means 2 audio channels.

- **A.** 2,646,000 bytes
- **B.** 5,292,000 bytes
- **C.** 21,168,000 bytes
- **D.** 42,336,000 bytes

Q26 (3 marks)

A simple 4x4 pixel black and white image (1-bit colour depth) is represented by the binary: 1100 0110 0110 1100. Describe what the image looks like.

- ○ **A.** A checkerboard pattern
- ○ **B.** A diagonal line from top-left to bottom-right
- ○ **C.** A diamond or cross shape in the centre
- ○ **D.** Horizontal stripes

Q27 (3 marks)

A photographer wants to reduce the file size of an image by half without changing the dimensions. Analyse which method would achieve this.

- ○ **A.** Double the colour depth from 12 bits to 24 bits
- ○ **B.** Reduce the colour depth from 24 bits to 12 bits
- ○ **C.** Double the number of pixels
- ○ **D.** Change the file name to be shorter

Q28 (3 marks)

A music producer has a 60-second audio file at 22,050 Hz sampling rate and 8-bit resolution. They want to improve the quality to CD standard (44,100 Hz, 16-bit). By what factor will the file size increase?

- ○ **A.** 2 times larger
- ○ **B.** 4 times larger
- ○ **C.** 8 times larger
- ○ **D.** 16 times larger

Q29 (3 marks)

A document contains 1,000 characters using only basic English letters and punctuation. Compare the file sizes if stored in 7-bit ASCII versus a Unicode format using 16 bits per character.

- ○ **A.** ASCII file is 7,000 bits; Unicode file is 16,000 bits; Unicode is approximately 2.3 times larger
- ○ **B.** Both files are the same size because they contain the same characters
- ○ **C.** ASCII file is 7,000 bytes; Unicode file is 16,000 bytes; Unicode is approximately 2.3 times larger
- ○ **D.** ASCII file is 8,000 bits; Unicode file is 16,000 bits; Unicode is exactly 2 times larger

Q30 (3 marks)

A website requires images to be under 50 kB. An image is 400 x 300 pixels. What is the maximum colour depth that can be used while staying under this limit?

- ○ **A.** 2 bits (4 colours)
- ○ **B.** 3 bits (8 colours)
- ○ **C.** 8 bits (256 colours)
- ○ **D.** 16 bits (65,536 colours)

Total: **55** marks • Time: **30 min**

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